Summary in Graph

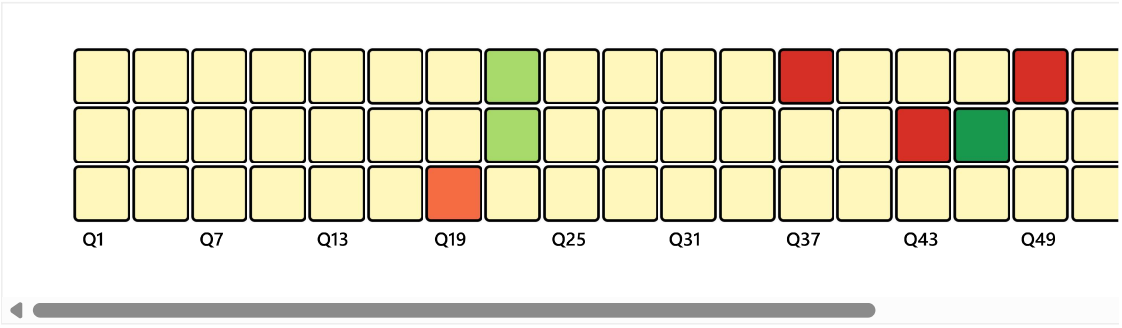
Exam Summary (GO Classes DA 2026 | All India Mock Test 4)

Qs. Attempted:	**14** (7 + 7)	**Correct Marks:**	**4** (2 + 2)
Correct Attempts:	**3** (2 + 1)	**Penalty Marks:**	**3** (0.33 + 2.67)
Incorrect Attempts:	**11** (5 + 6)	**Resultant Marks:**	**1** (1.67 + -0.67)

Total Questions:	**65** (30 + 35)
Total Marks:	**100** (30 + 70)
Exam Duration:	**180 Minutes**
Time Taken:	**180 Minutes**

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (23) out of 226 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Aptitude

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Quantitative Aptitude

Brother Santa and Chris walk to school from their house. The former takes 40 minutes while the latter, 30 minutes. One day Santa started 5 minutes earlier than Chris. In how many minutes would Chris overtake Santa?

A. 5

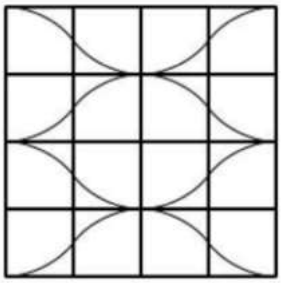
B. 15

C. 20

D. 25

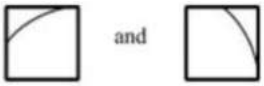
Your Answer: | Correct Answer: B | Not Attempted | Time taken: 06min 58sec | Discuss

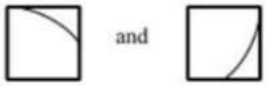
Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Spatial Aptitude

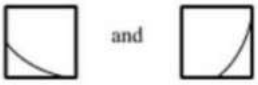


Which combination of square tiles shown below can produce the given pattern on the floor?

A. 

B. 

C. 

D. 

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Quantitative Aptitude

The sequence $\frac{1}{\log_3 2}, \frac{1}{\log_6 2}, \frac{1}{\log_{12} 2}, \frac{1}{\log_{24} 2} \cdots$ is in

A. Arithmetic progression (AP)

B. Geometric progression (GP)

C. Harmonic progression (HP)

D. None of these

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Verbal Aptitude

Amar, Akbar and Anthony are three friends, one of whom is a doctor, another is an engineer and the third is a professor. Amar is not an engineer. Akbar is the shortest. The tallest person is a doctor. The engineer's height is the geometric mean of the heights of the other two. Then which of the following is true?

A. Amar is a doctor and he is the tallest

B. Akbar is a professor and he is the tallest

C. Anthony is an engineer and he is shortest

D. Anthony is a doctor and he is the tallest

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Quantitative Aptitude

Jar **W** contains 40 white marbles and jar **B** contains 40 black marbles. 10 black marbles from jar **B** are transferred to jar **W** and mixed thoroughly. Now, 10 marbles are randomly selected from jar **W** and put back into jar **B**, so that each jar again contains 40 marbles.

The number of black marbles in jar **W**

A. would be equal to the number of white marbles in jar **B**

B. would be more than the number of white marbles in jar **B**

C. would be less than the number of white marbles in jar **B**

D. cannot be determined from the information given

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Quantitative Aptitude

Raju has 14 currency notes in his pocket consisting of only $\mathrm{Rs.}\ 20$ notes and $\mathrm{Rs.}\ 10$ notes. The total money value of the notes is $\mathrm{Rs.}\ 230$. The number of $\mathrm{Rs.}\ 10$ notes that Raju has is

A. 5

B. 6

C. 9

D. 10

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Quantitative Aptitude

There are n professors in a university. Most of them are very studious, and they like to own lots of books. Now the following facts about this group of professors have been learnt.

- No two professors own exactly the same number of books.
- Each professor owns strictly less than n books.
- No professor has exactly 200 books.

Given the above information, which of the following is **NOT** a possible value of n ?

A. 100

B. 199

C. 200

D. 201

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Quantitative Aptitude

I bought a shirt at 10% discount and sold it to a friend at a loss of 10%. If the friend paid me $\text{Rs. } 729.00$ for the shirt, what was the undiscounted price of the shirt?

A. $\text{Rs. } 900$

B. $\text{Rs. } 800$

C. $\text{Rs. } 1000$

D. $\text{Rs. } 911.25$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Analytical Aptitude

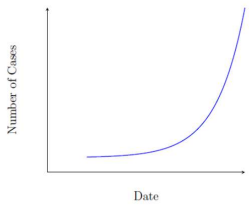
Following are the number of Coronavirus cases reported worldwide in the month of March 2021.

Date	Number of Cases
6 Mar	102050
17 Mar	198159
24 Mar	422574
26 Mar	531865

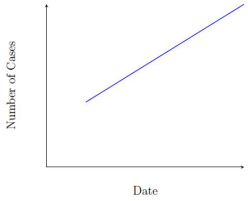
Which of the following graphs shows the best representation of the above data? Assume both axes are scaled linearly.

A.

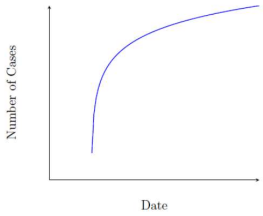
C.

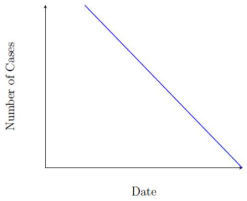


B.



D.





Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Verbal Aptitude

Talking to his wife over the telephone after their home was burgled, the husband said "Niether of the dogs _____ hurt, nor _____ any of the jewels stolen." Which of the following options is technically the most correct choice of verbs for the above blanks, in the order they would appear in the sentence above?

A. was, was

B. was, were

C. were, was

D. were, were

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Technical

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Let $S = \{\mathrm{u}_1, \ldots, \mathrm{u}_\mathrm{k}\}$ be a subset of non-zero vectors from $\mathbb{R}^n$. Now, consider the two statements given below:

I. If S is linearly dependent set in $\mathbb{R}^n$ then u_k is a linear combination of $\mathrm{u}_1, \ldots, \mathrm{u}_{\mathrm{k}-1}$.

II. If S is linearly independent set in $\mathbb{R}^n$ then $k < n$.

Which of the following statements is true?

A. Statement I is FALSE and Statement II is TRUE

B. Statement I is TRUE and Statement II is FALSE

C. Both Statement I and Statement II are FALSE

D. Both Statement I and Statement II are TRUE

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Let M be a 5×5 matrix with real entries such that $\text{Rank}(M) = 3$. Consider the linear system $Mx = b$. Let the row-reduced echelon form of the augmented matrix $[\,M \;\; b\,]$ be R, and let $R[i,:]$ denote the i-th row of R. Suppose that the linear system admits a solution. Which of the following statements is necessarily true?

A. $R[3,:] = [\,0 \;\; 1 \;\; 0 \;\; * \;\; * \;\; *\,]$

B. $R[5,:] = [\,0 \;\; 0 \;\; 1 \;\; 0 \;\; * \;\; *\,]$

C. $R[4,:] = [\,0 \;\; 0 \;\; 0 \;\; 1 \;\; * \;\; *\,]$

D. $R[4,:] = [\,0 \;\; 0 \;\; 0 \;\; 0 \;\; 0 \;\; 0\,]$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Suppose that a 3×3 matrix A has an eigenvalue -1. If the matrix

$$A + I = \begin{bmatrix} 1 & 0 & -2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix},$$

then the eigenvectors of A corresponding to the eigenvalue -1 are in the form,

A. $\begin{bmatrix} 2t \\ 0 \\ t \end{bmatrix}, \quad t \in \mathbb{R}$

B. $\begin{bmatrix} 2t \\ s \\ t \end{bmatrix}, \quad s, t \in \mathbb{R}$

C. $\begin{bmatrix} t \\ 0 \\ -2t \end{bmatrix}, \quad t \in \mathbb{R}$

D. $\begin{bmatrix} t \\ s \\ 2t \end{bmatrix}, \quad s, t \in \mathbb{R}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Let V be the column space of the matrix

$$A = \begin{pmatrix} 1 & -1 \\ 1 & 2 \\ 1 & -1 \end{pmatrix}.$$

Then the orthogonal projection of the vector $\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$ on V is

A. $[0\ 1\ 0]^T$

B. $[0\ 0\ 1]^T$

C. $[1\ 1\ 0]^T$

D. $[1\ 0\ 1]^T$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

The application of the Gram–Schmidt process of orthonormalization to the vectors

$$u_1 = (1,1,0), \quad u_2 = (1,0,0), \quad u_3 = (1,1,1)$$

yields the orthonormal set

A. $\frac{1}{\sqrt{2}}(1,1,0),\ (1,0,0),\ (0,0,1)$

B. $\frac{1}{\sqrt{2}}(1,1,0),\ \frac{1}{\sqrt{2}}(1,-1,0),\ \frac{1}{\sqrt{2}}(1,1,1)$

C. $(0,1,0),\ (1,0,0),\ (0,0,1)$

D. $\frac{1}{\sqrt{2}}(1,1,0),\ \frac{1}{\sqrt{2}}(1,-1,0),\ (0,0,1)$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 02min 50sec | Discuss

Q #6 | Multiple Select Type | Award: 1 | Penalty: 0 | Probability

Let A and B be two events.

Which of the following statements are **necessarily correct**?

A. $P(A \cup B) \leq P(A) + P(B)$

B. $P(A \cup B) \geq P(A) + P(B)$

C. $P(A \cap B) = P(A)\,P(B)$

D. $P(A \cap B) \geq P(A) + P(B) - 1$

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Select Type | Award: 1 | Penalty: 0 | Probability

For $n > 1$, let $X_1, X_2, \ldots, X_n$ be random variables such that

- $E(X_i) = 0$ for all i
- $E(X_i^2) = 1$ for all i
- $E(X_i X_j) = \rho$ for all $i \neq j$

Which of the following statements are true?

A. $\rho = 0$ if and only if $X_1, X_2, \ldots, X_n$ are independent

B. $\mathrm{Var}(X_1 + X_2 + \cdots + X_n) = n$ if and only if $X_1, X_2, \ldots, X_n$ are independent

C. $\mathrm{Var}(X_1 + X_2 + \cdots + X_n) = n$ if and only if $X_1, X_2, \ldots, X_n$ are pairwise independent

D. $\mathrm{Var}(X_1 + X_2 + \cdots + X_n) = n$ if and only if $\rho = 0$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Suppose X is a real-valued random variable. Which of the following values **CANNOT** be attained by $E[X]$ and $E[X^2]$, respectively?

A. 0 and 1

B. 2 and 3

C. $1/2$ and $1/3$

D. 2 and 5

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Suppose we learn decision trees from a training dataset D_{train} using a method that puts a limit on the maximum depth of the tree.

Two trees are learned:

- T_k: a tree with maximum depth k
- T_m: a tree with maximum depth m

Assume that $k < m$. After training, both trees are evaluated on the training set D_{train} and on a separate test set D_{test}.

Which of the following statements is most accurate?

A. T_k is likely to have larger error on both D_{train} and D_{test}.

B. T_m is likely to have larger error on both D_{train} and D_{test}.

C. T_k is likely to have larger error on D_{train}, but it is impossible to tell which tree will have larger error on D_{test}.

D. It is impossible to tell which tree has larger error on both D_{train} and D_{test}.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

What is the correct computation for the entropy of a dataset with 10 positive examples and 23 negative examples?

A. $-(10/33)\log(23/33) - (23/33)\log(10/33)$

B. $-(10/33)\log(10/33) - (23/33)\log(23/33)$

C. $-(10/23)\log(10/23) - (23/10)\log(23/10)$

D. $(10/33)\log(10/33) + (23/33)\log(23/33)$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Let a dataset $\mathcal{D}$ consist of N data points in D dimensions. Using PCA, the dataset is projected onto $d < D$ dimensions, resulting in a squared reconstruction error E.

Consider the following two statements:

- S_1: If an additional data point is added to the dataset and PCA is recomputed using the same target dimension d, then the new squared reconstruction error E' satisfies $E \leq E'$.
- S_2: If PCA is instead performed on the original dataset using $(d+1)$ dimensions, then the new squared reconstruction error E' satisfies $E \geq E'$.

Which one of the following choices is correct?

A. Only S_1 is true

B. Only S_2 is true

C. Both S_1 and S_2 are true

D. Neither S_1 nor S_2 is true

Your Answer: B | Correct Answer: C | Incorrect | Time taken: 06min 24sec | Discuss

Q #12 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Melbo is solving a binary classification problem with 1-D features. Features of datapoints labelled -1 are uniformly distributed in the interval $[0, 1]$, and features of datapoints labelled $+1$ are uniformly distributed in $[1.5, 2.5]$. That is,

$$\mathbb{P}[x \mid y = -1] = \mathcal{U}([0, 1]), \qquad \mathbb{P}[x \mid y = +1] = \mathcal{U}([1.5, 2.5]).$$

Melbo uses a threshold classifier $f_\eta(\cdot)$ that classifies a data point with feature x as $+1$ if $x \geq \eta$ and as -1 if $x < \eta$.

For which of the following ranges of η does Melbo's classifier achieve the minimum possible misclassification rate?

A. $\eta \in (0.5,\ 1]$

B. $\eta \in (1,\ 1.5]$

C. $\eta \in (1.5,\ 2]$

D. $\eta \in [0,\ 0.5)$

Your Answer: B | Correct Answer: B | Correct | Time taken: 39min 26sec | Discuss

Q #13 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

The function $f(x) := ax^2 + bx + c, a, b, c \in \mathbb{R}, a \neq 0$ with $\lim_{x\to\infty} f(x) = \infty$ has

A. a unique point of minimum in $\mathbb{R}$.

B. a unique point of maximum in $\mathbb{R}$.

C. exactly two points of minimum in $\mathbb{R}$.

D. exactly two points of maximum in $\mathbb{R}$

Your Answer: A | Correct Answer: A | Correct | Time taken: 05min 17sec | Discuss

Q #14 | Numerical Type | Award: 1 | Penalty: 0 | Calculus

Let

$$f(x) = 1 + x + \frac{x^2}{2} + \frac{x^3}{3} + \cdots + \frac{x^{2026}}{2026}.$$

Find the value of $f'(1)$.

Your Answer: | Correct Answer: 2026 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

An even function $f(x)$ has left derivative 5 at $x = 0$. Then

A. the right derivative of $f(x)$ at $x = 0$ need not exist

B. the right derivative of $f(x)$ at $x = 0$ exists and is equal to 5

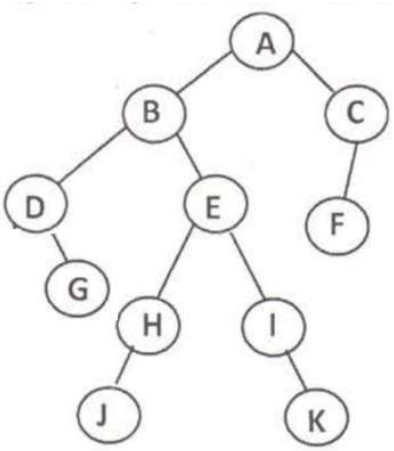
C. the right derivative of $f(x)$ at $x = 0$ exists and is equal to -5

D. none of the above is necessarily true

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

A node of a rooted binary tree is said to have preorder number i , if the node occurs in the ith position in the preorder traversal sequence. The notions of postorder number and inorder number are defined in a similar manner. We use the notations $\mathrm{PRE}(X), \mathrm{POST}(X)$ and IN (X) to denote the preorder number, postorder number and Inorder number of a node X respectively. Consider the following tree:



For the tree given above, which of the following is TRUE.

A. $(\mathrm{PRE}(\mathrm{A}), \mathrm{POST}(\mathrm{H}), \mathrm{IN}(\mathrm{F})) = (1, 4, 9)$

B. $(\mathrm{PRE}(\mathrm{B}), \mathrm{POST}(\mathrm{I}), \mathrm{IN}(\mathrm{D})) = (2, 6, 1)$

C. $(\mathrm{PRE}(\mathrm{C}), \mathrm{POST}(\mathrm{D}), \mathrm{IN}(\mathrm{E})) = (10, 3, 6)$

D. $(\mathrm{PRE}(\mathrm{J}), \mathrm{POST}(\mathrm{K}), \mathrm{IN}(\mathrm{G})) = (7, 5, 3)$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

Consider two sets of integers, $S = \{s_1, s_2, \ldots, s_m\}$ and $T = \{t_1, t_2, \ldots, t_n\}$. An algorithm uses a hash table to test whether $S \subseteq T$.

The algorithm first inserts all elements of T into a hash table, and then checks whether each element of S exists in the hash table.

Which of the following correctly describes the **average-case time complexity** of this algorithm?

A. $\Theta(n \log n + m \log n)$

B. $\Theta(n + m)$

C. $\Theta(nm)$

D. $\Theta(\log n + \log m)$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Assuming that the array contains random values, evenly distributed, consider the following function:

```
int find_max(const int *array, int n)
{
  int max = array[0];
  for (int i = 1; i < n; ++i)
  {
    if (array[i] > max)
    {
      max = array[i];
    }
  }
  return max;
}
```

On average, how many times does the variable `max` get assigned during the execution of the function?

A. $\Theta(\log n)$

B. $\Theta(\sqrt{n})$

C. $\Theta(n)$

D. $\Theta(1)$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Consider the three functions $f(n) = n^2 + n + 234, g(n) = 12n^2 + 34$ and $h(n) = 123n + 4$. Which of the following statements is not true?

A. $f(n) = \mathrm{O}(g(n))$

B. $g(n) = \mathrm{O}(f(n))$

C. $f(n) = \mathrm{O}(h(n))$

D. $h(n) = \mathrm{O}(f(n))$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 | Numerical Type | Award: 1 | Penalty: 0 | Programming in Python

Consider the following program segment.

```
x = 1
i = 1
while x <= 1000:
  x = 2 ** x
  i = i + 1
```

What is the value of `i` after execution of the program segment?

Your Answer: 4 | Correct Answer: 5 | Incorrect | Time taken: 51min 33sec | Discuss

Q #21 | Multiple Select Type | Award: 1 | Penalty: 0 | Artificial Intelligence

If $f(s), g(s)$ and $h(s)$ are all admissible heuristics then which of the following are also guaranteed to be admissible heuristics:

A. $f(s) + g(s) + h(s)$

B. $f(s)/6 + g(s)/3 + h(s)/2$

C. $\min(f(s), g(s), h(s))$

D. $\max(f(s), g(s), h(s))$

Your Answer: | Correct Answer: B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 | Multiple Select Type | Award: 1 | Penalty: 0 | Artificial Intelligence

Slowmo the snail is trying to find a path to a goal in an $N \times M$ grid maze. The available actions are to move one square up, down, left, right, and each action costs 1 . Moves into walls are impossible. The goal test is a black-box function (the snail does not know how the function works) that returns true if its state argument is a goal state, and false otherwise.

Assume that there may or may not be a reachable goal state. Which algorithms are complete for this problem? Select all that apply.

A. Breadth-first tree search

B. Depth-first tree search

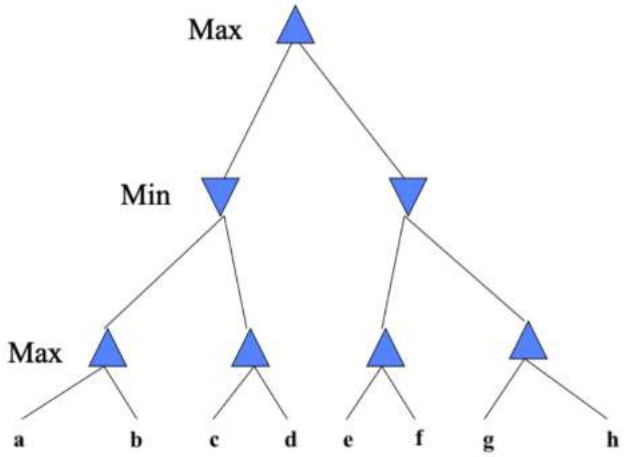
C. A* tree search with $h = 0$

D. A* graph search with $h = 0$

Your Answer: A;C;D | Correct Answer: D | Incorrect | Time taken: 02min 34sec | Discuss

Q #23 | Numerical Type | Award: 1 | Penalty: 0 | Artificial Intelligence

Consider the min-max tree given below, a to h represent numbers on the leaves. You should traverse the left sub-tree before the right sub-tree. What is the maximum number of leaves that can be pruned?



Your Answer: | Correct Answer: 3 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 | Multiple Select Type | Award: 1 | Penalty: 0 | Databases

Which of the following statements is/are true?

A. A weak entity set does not have any attribute that belongs to its key.

B. If a functional dependency $\mathrm{A} \rightarrow \mathrm{B}$ holds in relation $\mathrm{R}(\mathrm{A}, \mathrm{B}, \mathrm{C})$, then $\mathrm{AC} \rightarrow \mathrm{B}$ also holds.

C. If a relation R is in $3\mathrm{NF}, \mathrm{R}$ is also in BCNF.

D. If we define a foreign key in relation R, the DBMS checks the foreign-key constraint whenever a tuple in R is deleted.

Your Answer: A;D | Correct Answer: B | Incorrect | Time taken: 02min 41sec | Discuss

Q #25 | Numerical Type | Award: 1 | Penalty: 0 | Databases

Consider two relations $R(A, B)$ and $S(B, C)$. The number of tuples in R and S are $T(R) = 200$ and $T(S) = 100$, respectively. Attribute B is a key of relation R. What is the maximum value possible for $T(R \bowtie S)$?

Your Answer: 200 | Correct Answer: 100 | Incorrect | Time taken: 05min 14sec | Discuss

Q #26 | Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Let A be an $m \times n$ matrix. Define the following matrices obtained from A:

- $A(1, :)$: matrix obtained by deleting row 1 of A
- $A(:, 1)$: matrix obtained by deleting column 1 of A
- $A(1, 1)$: matrix obtained by deleting both row 1 and column 1 of A

Which of the following statements is/are true?

A. $(\operatorname{rank}(A)) - 2 \leq \operatorname{rank}(A(1,1)) \leq \operatorname{rank}(A)$

B. $\operatorname{rank}(A(1, :)) = \operatorname{rank}(A(:, 1))$

C. If $\operatorname{rank}(A) = \operatorname{rank}(A(1, :)) = \operatorname{rank}(A(:, 1))$, then $\operatorname{rank}(A) = \operatorname{rank}(A(1, 1))$

D. $\text{rank}(A(1,:)) + \text{rank}(A(:,1)) + 2 \geq 2\ \text{rank}(A)$

Your Answer: | Correct Answer: A;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Let A be an $n \times n$ matrix of rank 1. Let $\alpha = \det(I + A)$, where I is the identity matrix, and let $\beta = \text{trace}(A)$. Which of the following is true?

A. $\beta - \alpha = 1$

B. $\alpha - \beta = 1$

C. $\alpha < \beta + 1$

D. $\alpha > \beta + 1$

Your Answer: C | Correct Answer: B | Incorrect | Time taken: 04min 53sec | Discuss

Q #28 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Let us define a matrix $A \in M_n(\mathbb{R})$ to be *positive* if for every column vector $v \in \mathbb{R}^n$, we have $\langle Av, v \rangle \geq 0$, where $\langle \cdot, \cdot \rangle$ is the standard inner product on $\mathbb{R}^n$.

Let

$$A_{\alpha,\beta} = \begin{pmatrix} \alpha & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & \beta \end{pmatrix}.$$

Let

$$S = \{(\alpha, \beta) \in \mathbb{R}^2 : A_{\alpha,\beta} \text{ is positive}\}.$$

Which of the following statements is true?

A. S is empty

B. $(\alpha, \beta) \in S$ if and only if $\alpha\beta > 0$

C. $(\alpha, \beta) \in S$ if and only if $\alpha + \beta + 4 > 0$

D. $S = \mathbb{R}^2$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

The time between successive failures (in hours) of a side discharge loader operating in a mechanised underground coal mine are as follows:

$$62, 58, 54, 50, 52, 60, 58, 57, 50, 53$$

If the failure data follow an exponential distribution, then reliability of the equipment for a period of 50 hours is

A. 0.25

B. 0.40

C. 0.60

D. 1.00

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

A determinant is chosen at random from the set of all determinants of order 2 with elements 0 or 1 only. The probability of choosing a non-zero determinant is

A. $3/16$

B. $3/8$

C. $1/4$

D. none of these

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #31 | Numerical Type | Award: 2 | Penalty: 0 | Probability

Let X and Y be continuous random variables with joint probability density function

$$f(x,y) = \begin{cases} 4e^{-2y}, & \text{for } x \text{ positive and less than } y,\ y \text{ positive,} \\ 0, & \text{otherwise.} \end{cases}$$

Find the value of $E(X \mid Y = 2)$.

Your Answer: .146 | Correct Answer: 1 | Incorrect | Time taken: 06min 56sec | Discuss

Q #32 | Multiple Select Type | Award: 2 | Penalty: 0 | Probability

Let X and Y be random variables with cumulative distribution functions (CDFs) F and G, respectively.

Assume that F is continuous and, by the definition of a CDF, G is right-continuous (hence $G(-x)$ is left-continuous).

Which of the following functions of x are *necessarily* cumulative distribution functions (CDFs)?

A. $1 - G(-x) + F(x)$

B. $1 - F(-x) + G(x)$

C. $F(x)\,G(x)$

D. $(1 - F(x))(1 - G(x))$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #33 | Multiple Select Type | Award: 2 | Penalty: 0 | Probability

Let X_1, X_2 be i.i.d. random variables which are uniformly distributed on the interval $(0, \theta)$.

Consider the problem of testing

$$H_0 : \theta = 1 \quad \text{versus} \quad H_1 : \theta = 2.$$

Which of the following tests have size ≤ 0.5 and power ≥ 0.5?

A. Reject H_0 if and only if $\max\{X_1, X_2\} \geq 1.6$

B. Reject H_0 if and only if $\min\{X_1, X_2\} \geq 1$

C. Reject H_0 if and only if $X_1 \geq 0.4$

D. Reject H_0 if and only if $X_1 - X_2 \geq 0$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #34 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

The joint probability density function of the random variables X and Y is given by

$$f(x, y) = \begin{cases} \frac{6}{5}(x + y^2), & 0 \leq x \leq 1,\ 0 \leq y \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

Find the covariance $\text{Cov}(X, Y)$.

A. -0.01

B. 0

C. 0.01

D. 0.02

Your Answer: C | Correct Answer: A | Incorrect | Time taken: 12min 28sec | Discuss

Q #35 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

In an experiment, a fair die is rolled repeatedly until two sixes are obtained in succession. What is the probability that the experiment ends exactly on the fifth trial?

A. $125/6^5$

B. $150/6^5$

C. $175/6^5$

D. $200/6^5$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #36 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Suppose you used cross-validation to find a good value of the regularization parameter λ on a dataset with 500 training examples and 100 features. You now acquire more data, increasing the dataset size to $50{,}000$ examples (with the same features). Which of the following is most likely to work best, and why?

A. Use the same value of λ, since regularization depends only on the number of features

B. Use a higher value of λ, since more data increases the risk of overfitting

C. Use a lower value of λ, since variance decreases with more data

D. Use a higher value of λ, since bias should be increased with more data

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #37 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Consider a dataset with seven points $\{x_1, \ldots, x_7\}$. Given below are the distances between all pairs of points.

	x_1	x_2	x_3	x_4	x_5	x_6	x_7
x_1	0	5	3	1	6	2	3
x_2	5	0	4	6	1	7	8
x_3	3	4	0	4	3	5	6
x_4	1	6	4	0	7	1	2
x_5	6	1	3	7	0	8	9
x_6	2	7	5	1	8	0	1
x_7	3	8	6	2	9	1	0

Assume that $k = 2$, and the cluster centers are initialized to x_3 and x_6. Which of the following shows the two clusters formed at the end of the first iteration of k-means?

Circle the correct option.

A. $\{x_1, x_2, x_3, x_4\}, \{x_5, x_6, x_7\}$

B. $\{x_2, x_3, x_5\}, \{x_1, x_4, x_6, x_7\}$

C. $\{x_1, x_2, x_3, x_5\}, \{x_4, x_6, x_7\}$

D. $\{x_2, x_3, x_4, x_7\}, \{x_1, x_5, x_6\}$

Your Answer: B | Correct Answer: B | Correct | Time taken: 05min 35sec | Discuss

Q #38 | Numerical Type | Award: 2 | Penalty: 0 | Machine Learning

Imagine that you are given the following set of training examples. All the features are Boolean-valued.

$F1$	$F2$	$F3$	Result
T	T	F	$+$
F	T	T	$+$
T	F	T	$-$
F	T	F	$-$
F	F	T	$-$

Using a Naive Bayes classifier (without any smoothing), how will the following test example be classified?

$$F1 = T, \quad F2 = F, \quad F3 = F$$

Enter 1 for the positive class and -1 for the negative class.

Your Answer: | Correct Answer: -1 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #39 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

In this problem, we consider a variation of the Perceptron Learning Algorithm. Recall the perceptron algorithm given below.

$\text{Perceptron}(\{(x_i, y_i)\}_{i=1}^{T})$:

$\textcolor{gray}{1.}$ initialize w

$\textcolor{gray}{2.}$ for each $(x, y) \in \{(x_i, y_i)\}_{i=1}^{T}$:

  $\textcolor{gray}{3.}$ if $y(w^T x) \leq 0$:

  $\textcolor{gray}{4.}$ $w \leftarrow w + \eta y x$

$\textcolor{gray}{5.}$ return w

Which of the following choices of the learning rate η guarantees the following property?

If the algorithm encounters two consecutive occurrences of the same training example (x, y), then it will never make a mistake on the second occurrence.

Choose the correct option.

A. $\eta \geq \dfrac{1 - y(w^T x)}{\|x\|^2 + 1}$

B. $\eta \leq \dfrac{-y(w^T x)}{\|x\|^2 + 1}$

C. $\eta \leq \dfrac{-y(w^T x)}{\|x\|^2}$

D. $\eta \geq \dfrac{-y(w^T x)}{\|x\|^2}$

Your Answer: B | Correct Answer: D | Incorrect | Time taken: 00min 00sec | Discuss

Q #40 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

The regularized weight $\mathbf{w}_{\text{reg}}$ is a solution to:

$$\min_{\mathbf{w}} \frac{1}{N} \sum_{n=1}^{N} (\mathbf{w}^\top \mathbf{x}_n - y_n)^2 \quad \text{subject to} \quad \mathbf{w}^\top \lambda^\top \lambda \mathbf{w} \leq C,$$

where λ is a matrix. If $\mathbf{w}_{\text{lin}}^{\top}\lambda^{\top}\lambda\mathbf{w}_{\text{lin}} \leq C$, where $\mathbf{w}_{\text{lin}}$ is the linear regression solution, then what is $\mathbf{w}_{\text{reg}}$?

A. $\mathbf{w}_{\text{reg}} = \mathbf{w}_{\text{lin}}$

B. $\mathbf{w}_{\text{reg}} = \lambda\mathbf{w}_{\text{lin}}$

C. $\mathbf{w}_{\text{reg}} = \lambda^{\top}\lambda\mathbf{w}_{\text{lin}}$

D. $\mathbf{w}_{\text{reg}} = C\lambda\mathbf{w}_{\text{lin}}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #41 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Consider the following training set generated from a target function $f : \mathcal{Z} \to \{-1, +1\}$, where $\mathcal{Z} = \mathbb{R}^2$. The training data in the $\mathcal{Z}$-space are given directly as follows:

(z_1, z_2)	y
$(-3,\ 2)$	-1
$(0,\ -1)$	-1
$(0,\ 3)$	-1
$(1,\ 2)$	$+1$
$(3,\ -3)$	$+1$
$(3,\ 5)$	$+1$
$(3,\ 5)$	$+1$

Using geometry (not quadratic programming), what values of $\mathbf{w} = (w_1, w_2)$ and b specify the separating plane $\mathbf{w}^{\top}\mathbf{z} + b = 0$ that maximizes the margin in the $\mathcal{Z}$-space? The values of w_1, w_2, b are:

A. $-1,\ 1,\ -0.5$

B. $1,\ -1,\ -0.5$

C. $1,\ 0,\ -0.5$

D. $0,\ 1,\ -0.5$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #42 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

The number of saddle points of the function $f(x, y) = 2x^4 - x^2 + 3y^2$ is

A. 1

B. 0

C. 2

D. none of the above

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #43 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

If $\lim_{x \to 0} \frac{ae^x - b\cos x}{x} = 5$, then.

A. a and b are uniquely determined.

B. a is uniquely determined, but not b.

C. b is uniquely determined, but not a.

D. neither a nor b is uniquely determined.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #44 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

In deletion by merging in a Binary Search Tree, when a node with two children is deleted, the right sub-tree of the deleted node may be attached to the node containing largest key in the left sub-tree. If the node 15 is deleted from the tree in $\text{Figure } 1$, we obtain a structure as shown in $\text{Figure } 2$.

The values in the nodes marked X and Y in $\text{Figure } 2$ are

FIGURE 1

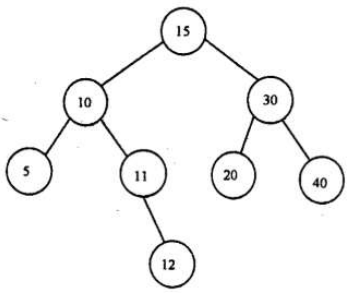
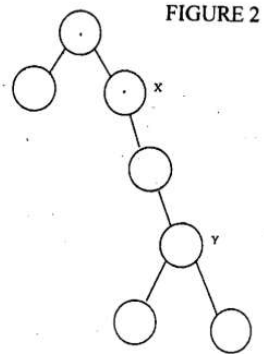


FIGURE 2



A. $\text{X} = 5,\ \text{Y} = 10.$

B. $\text{X} = 11,\ \text{Y} = 30.$

C. $\text{X} = 10,\ \text{Y} = 20.$

D. $\text{X} = 20,\ \text{Y} = 30.$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #45 | Numerical Type | Award: 2 | Penalty: 0 | Data Structures

Consider a special queue called **MirrorQueue** that supports three operations:

- `Enqueue(x)` — inserts an element at the rear
- `Dequeue()` — removes an element from the front
- `Front()` — returns the front element without removing it

Initially, MirrorQueue contains elements `[1, 2, 3, 4, 5]` (front $= 1$).

We want to **print all elements twice** in the following order:

$$1, 2, 3, 4, 5, 5, 4, 3, 2, 1$$

so that the **final content of the queue remains unchanged.**

You can only use the three supported operations (`Enqueue` , `Dequeue` , and `Front`).

What is the **minimum total number of operation calls** required?

Your Answer: 30 | Correct Answer: 50 | Incorrect | Time taken: 14min 14sec | Discuss

Q #46 | Multiple Select Type | Award: 2 | Penalty: 0 | Data Structures

Consider a depth-first search (DFS) performed on a directed graph $G = (V, E)$. Recall that in DFS of a directed graph, the presence of a **back edge** indicates the existence of a cycle.

Which of the following statements are **TRUE**?

A. If a DFS on G produces exactly one back edge, then it is possible to remove a single edge from G such that the resulting graph is acyclic.

B. If a directed graph G can be made acyclic by removing exactly one edge, then any DFS of G must encounter exactly one back edge.

C. A directed graph may contain multiple cycles but only one back edge during a DFS traversal.

D. Removing an arbitrary edge that lies on a cycle is always sufficient to make a directed graph acyclic.

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #47 | Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

Let $G = (V, E)$ be a directed, weighted graph with source s and destination t.

A new graph G' is formed by increasing the weight of every edge leaving s by a constant $C = 2026$.

Which of the following statements is/are TRUE?

A. The shortest path from s to t in G always remains unchanged in G'.

B. The shortest distance from s to t in G' is exactly C more than the shortest distance in G.

C. The cost of any $s - t$ path in G' increases by nC, where n is the number of outgoing edges of s.

D. The shortest path from s to t in G' may differ from that in G.

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #48 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Programming in Python

Let `n`, `i` and `total` be integer variables. The user enters a positive value of `n`. Which of the following Python program segments prints the largest value of `total`?

A.
```
total = 0
i = 1
while (i + 1) < n:
  i += 1
  total += i
print(total)
```

B.
```
total = 0
i = 1
while i < n:
  total += i
  i += 1
print(total)
```

C.
```
total = 0
for i in range(1, n):
  total += i
print(total)
```

D.
```
total = 0
for i in range(1, n + 1):
  total += i
print(total)
```

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #49 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Programming in Python

Two sequences $U(n)$ and $V(n)$ are mutually recursively defined as:

$$U(0) = 0, \quad U(n) = V(n-1) + 5 \quad \text{for } n \geq 1,$$
$$V(0) = 1, \quad V(n) = 2U(n-1) + 2 \quad \text{for } n \geq 1.$$

Complete the following Python code for recursively computing `U(n)` for an integer `n ≥ 0`. You must **not** use a separate function for computing `V(n)`.

```
def U(n):
  if n == 0: return 0
  if n == 1: return 6
  return __________ ;
```

Which of the following correctly replaces the blank?

A. `2 * U(n - 1) + 7`

B. `U(n - 1) + 2 * U(n - 2) + 5`

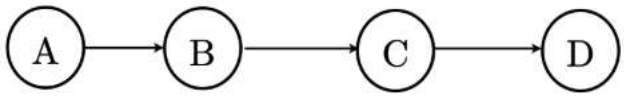
C. `2 * U(n - 2) + 7`

D. `2 * U(n - 1) + 5`

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #50 | Multiple Select Type | Award: 2 | Penalty: 0 | Artificial Intelligence

Assume the following Bayes' net, and the corresponding distributions over the variables in the Bayes' net:



$P(A)$	
$-a$	$3/4$
$+a$	$1/4$

$P(B\|A)$		
$-a$	$-b$	$2/3$
$-a$	$+b$	$1/3$
$+a$	$-b$	$4/5$
$+a$	$+b$	$1/5$

$P(C\|B)$		
$-b$	$-c$	$1/4$
$-b$	$+c$	$3/4$
$+b$	$-c$	$1/2$
$+b$	$+c$	$1/2$

$P(D\|C)$		
$-c$	$-d$	$1/8$
$-c$	$+d$	$7/8$
$+c$	$-d$	$5/6$
$+c$	$+d$	$1/6$

Recall that during Gibbs Sampling, samples are generated through an iterative process.
Assume that the only evidence that is available is $\mathrm{A} = +\mathrm{a}$. Which of the sequence(s) below could have been generated by Gibbs Sampling

A. Sequence 1

1 :	$+a$	$-b$	$-c$	$+d$
2 :	$+a$	$-b$	$-c$	$+d$
3 :	$+a$	$-b$	$+c$	$+d$

B. Sequence 2

1	$+a$	$-b$	$-c$	$+d$
2	$+a$	$-b$	$-c$	$-d$
3	$-a$	$-b$	$-c$	$+d$

C. Sequence 3

1	$+a$	$-b$	$-c$	$+d$
2	$+a$	$-b$	$-c$	$-d$
3	$+a$	$+b$	$-c$	$-d$

D. Sequence 4

1	$+a$	$-b$	$-c$	$+d$
2	$+a$	$-b$	$-c$	$-d$
3	$+a$	$+b$	$-c$	$+d$

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #51 | Multiple Select Type | Award: 2 | Penalty: 0 | Databases

Given two relations R and S with the same set of attributes, select all the equality equations that hold for every instance of R and S. We here assume the set semantics.

A. $\mathrm{R} \cap \mathrm{S} = \mathrm{R} - (\mathrm{R} - \mathrm{S})$

B. $\mathrm{R} \bowtie \mathrm{S} = \mathrm{S} \bowtie \mathrm{R}$

C. $\mathrm{R} \cap \mathrm{S} = \mathrm{R} \bowtie \mathrm{S}$

D. $\mathrm{R} \cap \mathrm{S} = \mathrm{R} \times \mathrm{S}$

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #52 | Numerical Type | Award: 2 | Penalty: 0 | Databases

Consider a $\mathrm{B}+$ Tree of order $d = 2$ (Note: Order is defined as maximum pointers = $2d + 1$, minimum entries = d).

- Max pointers per node = 5.
- Max keys per node = 4.
- Min keys per node = 2.

We insert the following keys in this specific order into an initially empty tree:

$$1, 2, 3, 4, 5, 6, 7, 8, 9, 10$$

However, the insertion strategy is Standard Insertion (splitting nodes as they fill), NOT "Bulk Loading".

How many node splits (including leaf and internal nodes) occur during this entire process?

Your Answer: | Correct Answer: 3 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #53 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

Consider a relation schema

$$\mathrm{R(A, B, C, D, E)}$$

With the following set of functional dependencies:

$$F = \{\mathrm{A} \rightarrow \mathrm{B}, \mathrm{C} \rightarrow \mathrm{D}, \mathrm{AC} \rightarrow \mathrm{E}\}$$

A student proposes the following decomposition of R :

$$\mathrm{R1(\ A,\ B), R2(C, D), R3(\ A, C, E)}$$

Which of the following statements about the above decomposition is CORRECT?

A. The decomposition is lossless join and dependency preserving.

B. The decomposition is lossless join but not dependency preserving.

C. The decomposition is dependency preserving but not lossless join.

D. The decomposition is neither lossless join nor dependency preserving.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #54 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Artificial Intelligence

Consider the following statements where $A = \{0,1\},\ B = \{-1,1\},\ C = \{-1,0,1\}$:

(i) $\forall x \in A\ \forall y \in B\ \forall z \in C\ \big((x < y) \rightarrow (y < z)\big)$.

(ii) $\exists x \in A\ \exists y \in B\ \exists z \in C\ \big((x < y) \wedge (y < z)\big)$.

(iii) $\forall x \in A\ \exists y \in B\ \big((x^2 \leq y^2) \rightarrow \forall z \in C\ (y \leq z)\big)$.

Which of the above statements are true?

A. None of options (A), (B) , (C), (D) is correct.

B. Only (iii).

C. Only (i) and (ii).

D. Only (ii) and (iii).

Your Answer: C | Correct Answer: B | Incorrect | Time taken: 12min 53sec | Discuss

Q #55 | Multiple Select Type | Award: 2 | Penalty: 0 | Artificial Intelligence

Which of the following statements is/are false?

A. $(p \wedge q) \rightarrow r \equiv (p \rightarrow r) \wedge (q \rightarrow r)$

B. $(p \vee q) \rightarrow r \equiv (p \rightarrow r) \vee (q \rightarrow r)$

C. $p \wedge (q \rightarrow r) \equiv (p \wedge q) \rightarrow (p \wedge r)$

D. $p \vee (q \rightarrow r) \equiv (p \vee q) \rightarrow (p \vee r)$

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

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